Assignment 1: Analysis of Sorting Algorithms
By Group : Refer Lab 2 (assigned members)

Submission: Before or at 18.11.2024
(*Softcopy – capture from handwritten is the BEST*)

Objective
In this assignment, your group will implement and compare four different sorting algorithms using an array of student marks. You will track the number of **comparisons** and **swaps** each algorithm performs and analyze their efficiency.

Instructions
1. Group Formation:
   - Assign each group member a specific sorting algorithm:
     - Member 1: Bubble Sort
     - Member 2: Improved Bubble Sort (with optimization)
     - Member 3: Selection Sort
     - Member 4: Insertion Sort

2. Data Sample:
   - You will be given an array of 5 student marks (e.g., [75, 95, 60, 88, 70]).
   - Your task is to sort the marks in ascending order.

3. Tasks:
   - Implement the assigned sorting algorithm in C++.
   - Count and record the total number of comparisons and swaps performed by your algorithm. You can use C++ code and write manually to represent the comparison and swapping activity in each sorting technique.
   - Each member will complete their implementation individually but collaborate on analyzing and comparing results.

4. Requirements:
   - Write your code in C++. (Students can use lecture slide coding)
   - Use functions for sorting and tracking comparisons and swaps.
   - Ensure your code is well-commented and properly indented for readability.

Expected Output
- The sorted list of student marks.
- The total number of comparisons and swaps for each sorting algorithm.

Deliverables
1. Source Code:
   - Submit the C++ code for each sorting algorithm, clearly indicating the algorithm name in the file.

2. Analysis Report (Group Report):
   - Compare the performance (number of comparisons and swaps) of each algorithm.

- Discuss the efficiency of each algorithm based on your findings.
- Include a table summarizing your results (e.g., comparisons and swaps for each algorithm).

Sample Output Format (for the Report)

Sorting Algorithm	Comparisons	Swaps
Bubble Sort	X	Y
Improved Bubble Sort	X	Y
Selection Sort	X	Y
Insertion Sort	X	Y

Sample C++ Code Structure (for Bubble Sort)

Submission Deadline
- The assignment is due in one week from the assignment date. Ensure that all deliverables are submitted on time.

This assignment not only reinforces the understanding of sorting algorithms but also encourages collaboration, analytical thinking, and communication skills.